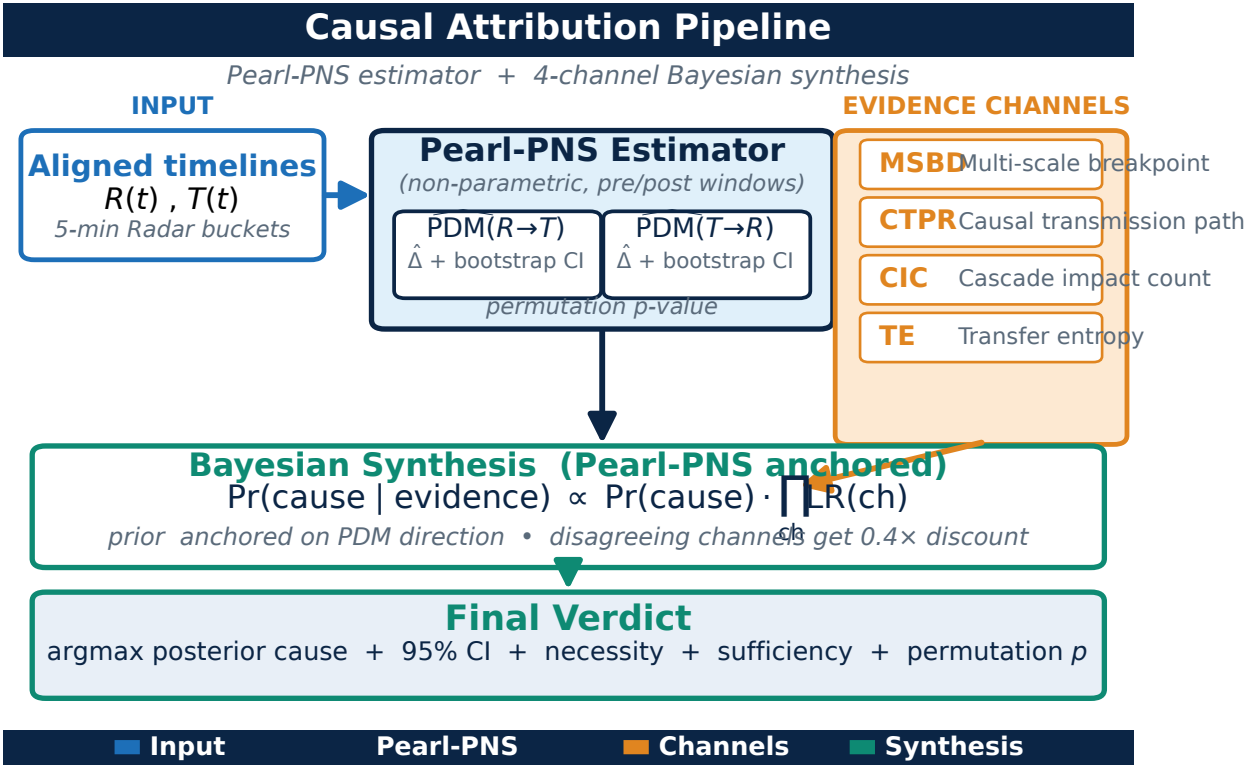
Causal Attribution Pipeline
Pearl-PNS estimator + 4-channel Bayesian synthesis
INPUT
Aligned timelines
R(t) , T(t)
5-min Radar buckets
Pearl-PNS Estimator
(non-parametric, pre/post windows)
PDM(R→T)
Δ̂ + bootstrap CI
PDM(T→R)
Δ̂ + bootstrap CI
permutation p-value
EVIDENCE CHANNELS
MSBD Multi-scale breakpoint
CTPR Causal transmission path
CIC Cascade impact count
TE Transfer entropy
Bayesian Synthesis (Pearl-PNS anchored)
Pr(cause | evidence) ∝ Pr(cause) · ∏ LR(ch)
ch
prior anchored on PDM direction • disagreeing channels get 0.4× discount
Final Verdict
argmax posterior cause + 95% CI + necessity + sufficiency + permutation p
Input
Pearl-PNS
Channels
Synthesis